Simultaneous Equations and Inequalities

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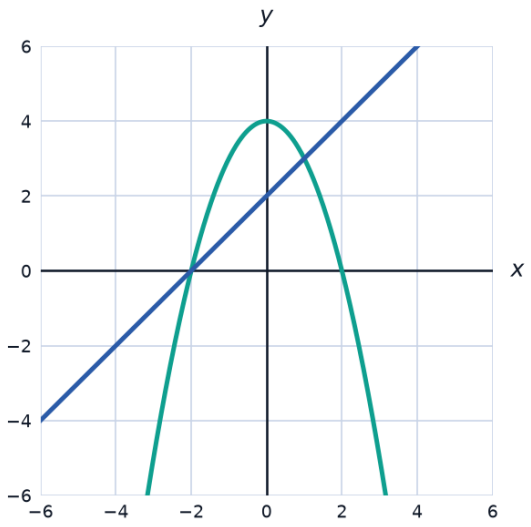
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 9$. What is the value of x?

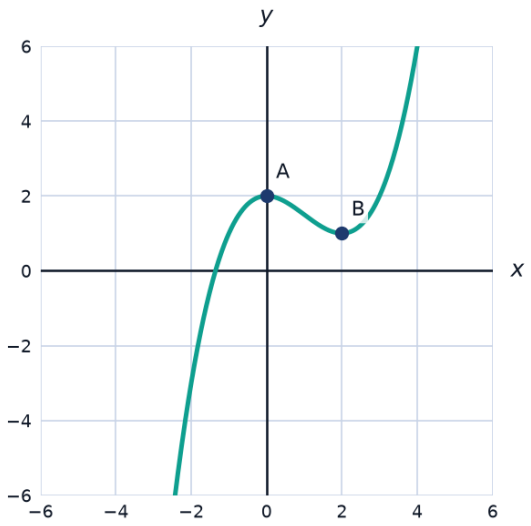
2 [Foundational] • Target: Linear-Quadratic Simultaneous Systems

A curve has equation $y = \frac{kx}{2x-1}$, where $x \neq \frac{1}{2}$ and k is a non-zero constant. The normal to the curve at the point where $x = 1$ has a gradient of $\frac{1}{3}$. Given that x is increasing at a rate of 0.6 units per second, find the rate of change of y when $x = 2$.



3 [Foundational] • Target: Quadratic Inequalities in One Variable

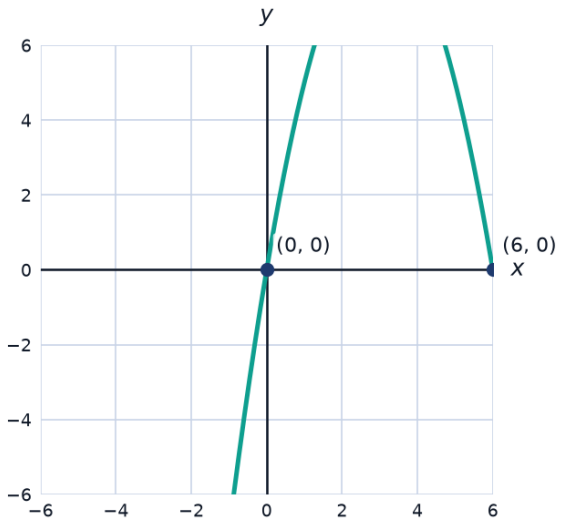
Solve the logarithmic equation $\log_2(5x + 6) - \log_2(x) = 3$.



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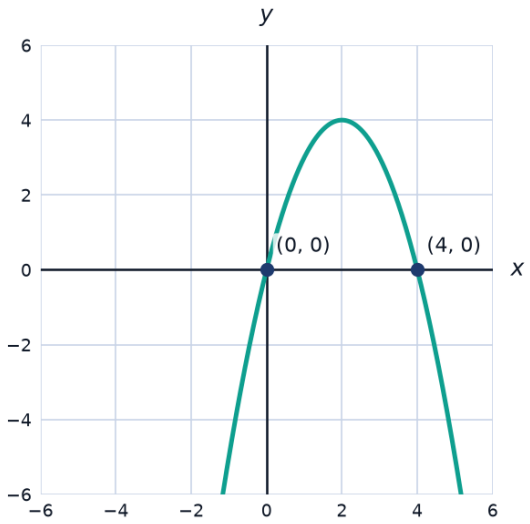
4 [Foundational] • Target: Rational Inequalities and Sign Analysis

The diagram shows part of the curve $y = 6x - x^2$, which intersects the x-axis at $(0, 0)$ and $(6, 0)$. Find the area of the region enclosed between the curve and the x-axis.



5 [Foundational] • Target: Systems with Modulus and Geometric Applications

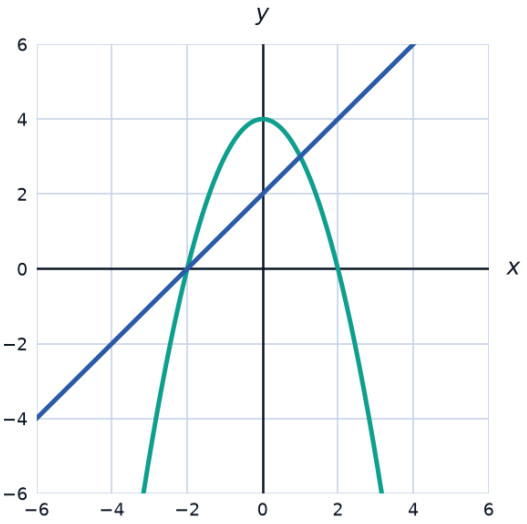
The diagram shows the curve $y = 4x - x^2$. Find the area of the region enclosed by the curve and the x-axis.



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6 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the exponential equation $2^{x+2} = 5^x$, giving x in terms of common logarithms.



7 [Foundational] • Target: Linear-Quadratic Simultaneous Systems

The variables x and y are connected by the equation $y = \sqrt{2x+5}$. Given that x increases at a constant rate of 6 units per second, find the rate of change of y with respect to time when $x = 2$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Quadratic Inequalities in One Variable

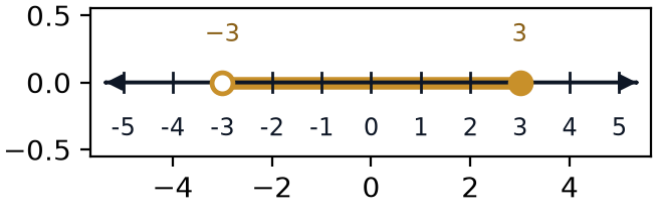
The polynomial $P(x) = 2x^3 + ax^2 + bx - 6$ has a factor of $x - 2$ and leaves a remainder of -24 when divided by $x + 1$, where a and b are constants. Find the remainder when $P(x)$ is divided by $x - 3$.

9 [Intermediate] • Target: Rational Inequalities and Sign Analysis

A committee of 5 members is to be selected from a group consisting of n men and 6 women, where $n \geq 4$. Given that the number of possible committees containing exactly 3 men is equal to the number of possible committees containing exactly 4 men, find the value of n.

10 [Intermediate] • Target: Systems with Modulus and Geometric Applications

In the expansion of $(1 + ax)^n$ in ascending powers of x, where n is a positive integer and a is a constant, the coefficient of x is -12 and the coefficient of x^2 is 60. Find the coefficient of x^3.



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11 [Intermediate] • Target: Simultaneous Equations & Inequalities Practice Exercises

Relative to a fixed origin O, ship A has an initial position vector of $(24\mathbf{i} + 10\mathbf{j})$ km and moves with a constant velocity of $(-2\mathbf{i} + 5\mathbf{j})$ km h^{-1}. Ship B has an initial position vector of $(-20\mathbf{i} - 18\mathbf{j})$ km and moves with a constant velocity of $(u\mathbf{i} + 12\mathbf{j})$ km h^{-1}, where u is a constant. Given that ship B intercepts ship A, find the speed of ship B.

12 [Intermediate] • Target: Linear-Quadratic Simultaneous Systems

In triangle ABC, $AB = 6$ cm, $AC = 10$ cm, and the area of triangle ABC is $15\sqrt{3}$ cm^2. Given that angle BAC is obtuse, find the exact length of BC.

13 [Intermediate] • Target: Quadratic Inequalities in One Variable

The variables x and y are related by the equation $y = 2x^3 - 5x$. Given that x is increasing at a constant rate of 3 units per second, find the rate of change of y with respect to time at the instant when $x = 2$.

14 [Intermediate] • Target: Rational Inequalities and Sign Analysis

Find the real solution to the equation $\log_3 x + \log_3 (x - 8) = 2$.

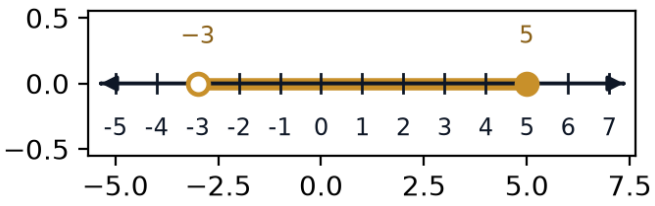
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Systems with Modulus and Geometric Applications • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Find the remainder when the polynomial $f(x) = 4x^3 - 6x^2 + 3x - 1$ is divided by $2x - 1$.

16 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

The position vectors of points A and B relative to an origin O are given by $\vec{OA} = 5\mathbf{i} - 2\mathbf{j}$ and $\vec{OB} = -\mathbf{i} + 6\mathbf{j}$. Find the magnitude of $\vec{AB}$.



17 [Challenging] • Target: Linear-Quadratic Simultaneous Systems • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

Find the value of the integer $n \geq 3$ that satisfies the equation ${}^nP_3 = 10 \times {}^nC_2$

18 [Challenging] • Target: Quadratic Inequalities in One Variable • Cross-Lesson: 6.01 Permutations and Combinations

Given that ${}^nP_2 = n3$, where n is an integer such that $n \geq 3$, find the value of n.

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19 [Challenging] • Target: Rational Inequalities and Sign Analysis • Cross-Lesson: 7.01 Differentiation and Rates of Change

When the polynomial $P(x) = 2x^3 - 3x^2 + ax - 8$ is divided by $x - 3$, the remainder is 10. What is the value of the constant a?

20 [Challenging] • Target: Systems with Modulus and Geometric Applications • Cross-Lesson: 8.01 Integration and Kinematics

The position vectors of points A and B relative to a fixed origin O are given by $\vec{OA} = -2\mathbf{i} + 5\mathbf{j}$ and $\vec{OB} = 4\mathbf{i} - 3\mathbf{j}$. Find the magnitude of the vector $\vec{AB}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the modulus equation \(|2x + 3| = 7 \).

a. \(x = 3 \) and \(x = -3 \)

b. \(x = -2 \) and \(x = -\frac{10}{2} \)

c. \(x = 2 \) only

d. \(x = 2 \) and \(x = -\frac{10}{2} \)

22 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the modulus equation \(|3x + 3| = 9 \).

a. \(x = 2 \) and \(x = -\frac{12}{3} \)

b. \(x = 3 \) and \(x = -3 \)

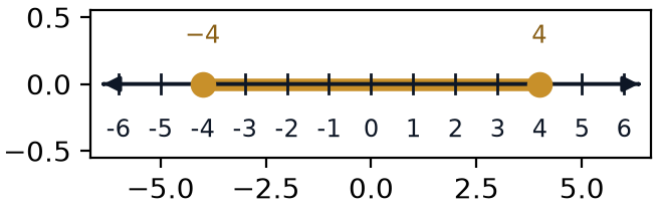
c. \(x = -2 \) and \(x = -\frac{12}{3} \)

d. \(x = 2 \) only

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23 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the modulus equation \(|4x + 3| = 11 \).



a. \(x = 2 \) only

b. \(x = 3 \) and \(x = -3 \)

c. \(x = 2 \) and \(x = -\frac{14}{4} \)

d. \(x = -2 \) and \(x = -\frac{14}{4} \)

24 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the modulus equation \(|5x + 3| = 13 \).

a. \(x = 2 \) and \(x = -\frac{16}{5} \)

b. \(x = 2 \) only

c. \(x = -2 \) and \(x = -\frac{16}{5} \)

d. \(x = 3 \) and \(x = -3 \)

25 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the modulus equation \(|6x + 3| = 15 \).

a. \(x = 3 \) and \(x = -3 \)

b. \(x = 2 \) and \(x = -\frac{18}{6} \)

c. \(x = 2 \) only

d. \(x = -2 \) and \(x = -\frac{18}{6} \)

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26 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the modulus equation $|7x + 3| = 17$.

a. $x = 2$ and $x = -\frac{20}{7}$

b. $x = 2$ only

c. $x = -2$ and $x = -\frac{20}{7}$

d. $x = 3$ and $x = -3$

27 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the modulus equation $|8x + 3| = 19$.

a. $x = 2$ and $x = -\frac{22}{8}$

b. $x = 2$ only

c. $x = -2$ and $x = -\frac{22}{8}$

d. $x = 3$ and $x = -3$

28 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the modulus equation $|9x + 3| = 21$.

a. $x = 2$ only

b. $x = 2$ and $x = -\frac{24}{9}$

c. $x = 3$ and $x = -3$

d. $x = -2$ and $x = -\frac{24}{9}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Linear-Quadratic Simultaneous Systems

Solve the simultaneous system $y - 2x = 3$ and $x^2 + y^2 = 26$ for the positive solution pair (x, y).

a. $(1, 5)$

b. $(2, 7)$

c. $(3, 9)$

d. $(4, 11)$

30 [Intermediate] • Target: Linear-Quadratic Simultaneous Systems

Find the two points of intersection of the line $2x + y = 7$ and the rectangular hyperbola $xy = 6$.

a. $(2, 3)$ and $\left(\frac{3}{2}, 4\right)$

b. $(1, 5)$ and $(3, 1)$

c. $(2, 3)$ and $(3, 2)$

d. $(4, -1)$ and $(1, 6)$

31 [Intermediate] • Target: Linear-Quadratic Simultaneous Systems

Determine the value of k for which the line $x + y = k$ is tangent to the curve $y = \frac{4}{x}$.

a. $k = \pm 4$

b. $k = 4$ only

c. $k = \pm 2$

d. $k = 8$

32 [Intermediate] • Target: Quadratic Inequalities in One Variable

Find the set of values of x satisfying the quadratic inequality $2x^2 - 5x - 3 \geq 0$.

a. $x \leq -\frac{1}{2}$ or $x \geq 3$

b. $-\frac{1}{2} \leq x \leq 3$

c. $x < -1$ or $x > 3$

d. $x \geq 3$ only

33 [Intermediate] • Target: Quadratic Inequalities in One Variable

Solve the strict quadratic inequality $x(x + 4) < 12$.

a. $-6 < x < 2$

b. $x < -6$ or $x > 2$

c. $-2 < x < 6$

d. $0 < x < 3$

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34 [Intermediate] • Target: Quadratic Inequalities in One Variable

Find the range of values of x for which the expression $9 - x^2$ is strictly positive.

a. $-3 < x < 3$

b. $x < -3$ or $x > 3$

c. $x < 3$

d. $0 < x < 9$

35 [Intermediate] • Target: Rational Inequalities and Sign Analysis

Solve the rational inequality $\frac{x-2}{x+3} \leq 0$.

a. $-3 < x \leq 2$

b. $-3 \leq x \leq 2$

c. $x < -3$ or $x \geq 2$

d. $x \leq 2$

36 [Intermediate] • Target: Rational Inequalities and Sign Analysis

Find the complete solution set for the inequality $\frac{2x+1}{x-1} > 1$.

a. $x < -2$ or $x > 1$

b. $-2 < x < 1$

c. $x > 1$ only

d. $x > -2$

37 [Intermediate] • Target: Rational Inequalities and Sign Analysis

Solve the inequality $\frac{1}{x} < 2$ for all non-zero real x.

a. $x < 0$ or $x > \frac{1}{2}$

b. $0 < x < \frac{1}{2}$

c. $x > \frac{1}{2}$ only

d. $x < \frac{1}{2}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Systems with Modulus and Geometric Applications • Cross-Lesson: 6.01 Permutations and Combinations

Find the values of x satisfying the simultaneous conditions $|x - 3| \leq 4$ and $x^2 - 4x > 0$.

a. $-1 \leq x < 0$ or $4 < x \leq 7$

b. $-1 \leq x \leq 7$

c. $0 < x < 4$

d. $4 < x \leq 7$ only

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39 [Challenging] • Target: Systems with Modulus and Geometric Applications • Cross-Lesson: 7.01 Differentiation and Rates of Change

The perimeter of a right-angled triangle is 30 cm and its hypotenuse is 13 cm. Find the lengths of the two legs.

a. 5 cm and 12 cm

b. 6 cm and 11 cm

c. 7 cm and 10 cm

d. 8 cm and 9 cm

40 [Challenging] • Target: Systems with Modulus and Geometric Applications • Cross-Lesson: 8.01 Integration and Kinematics

Find the range of values of x where the line $y = 3x - 2$ lies entirely above the parabola $y = x^2 - 2x + 4$.

a. $2 < x < 3$

b. $x < 2$ or $x > 3$

c. $-3 < x < -2$

d. No such interval exists

41 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

The diagram shows the graphs of the linear equation $y = 2x + c$ and the quadratic equation $y = -x^2 + 4x + 5$. The two graphs intersect at the points where $x = -1$ and $x = k$. Find the value of c and the value of k.

a. c = 3, k = 4

b. c = 5, k = 3

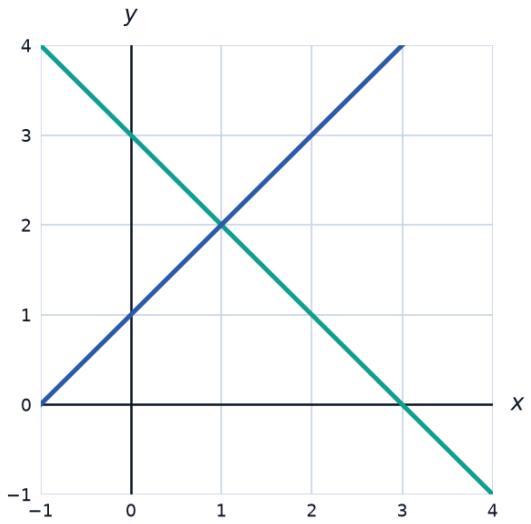
c. c = 3, k = 3

d. c = 5, k = 4

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42 [Challenging] • Target: Linear-Quadratic Simultaneous Systems • Cross-Lesson: 3.01 Indices, Surds and Polynomials

The diagram shows the graphs of two linear equations. The point of intersection of the two lines represents the solution to the system of simultaneous equations. Find the coordinates of the point of intersection.



a. (1, 2)

b. (2, 1)

c. (3, 0)

d. (0, 3)

43 [Challenging] • Target: Quadratic Inequalities in One Variable • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

Solve the inequality $x^2 + 9x + 20 < 0$.

a. $x - 4$

b. $-5 \leq x \leq -4$

c. -5<x<-4

d. $x4$

44 [Challenging] • Target: Rational Inequalities and Sign Analysis • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

Solve the inequality $x^2 + 4x + 3 < 0$.

a. $x1$

b. -3<x<-1

c. $-3 \leq x \leq -1$

d. $x - 1$

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45 [Challenging] • Target: Systems with Modulus and Geometric Applications • Cross-Lesson: 6.01 Permutations and Combinations

Solve the inequality $x^2 - x - 6 < 0$.

a. $-2 \leq x \leq 3$

b. $x - 3$

c. -2<x<3

d. $x3$

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